CaffeinatedData

ARIMA modeling

Research Question:

"Can past daily revenue data be used to forecast future revenue trends for the organization using time series modeling techniques?"

Project Goal:

The goal of this data analysis is to examine historical daily revenue data and develop a univariate time series forecasting model capable of predicting future revenue trends. The analysis will identify patterns such as trends, seasonality, and autocorrelation within the revenue data to support more accurate forecasting.

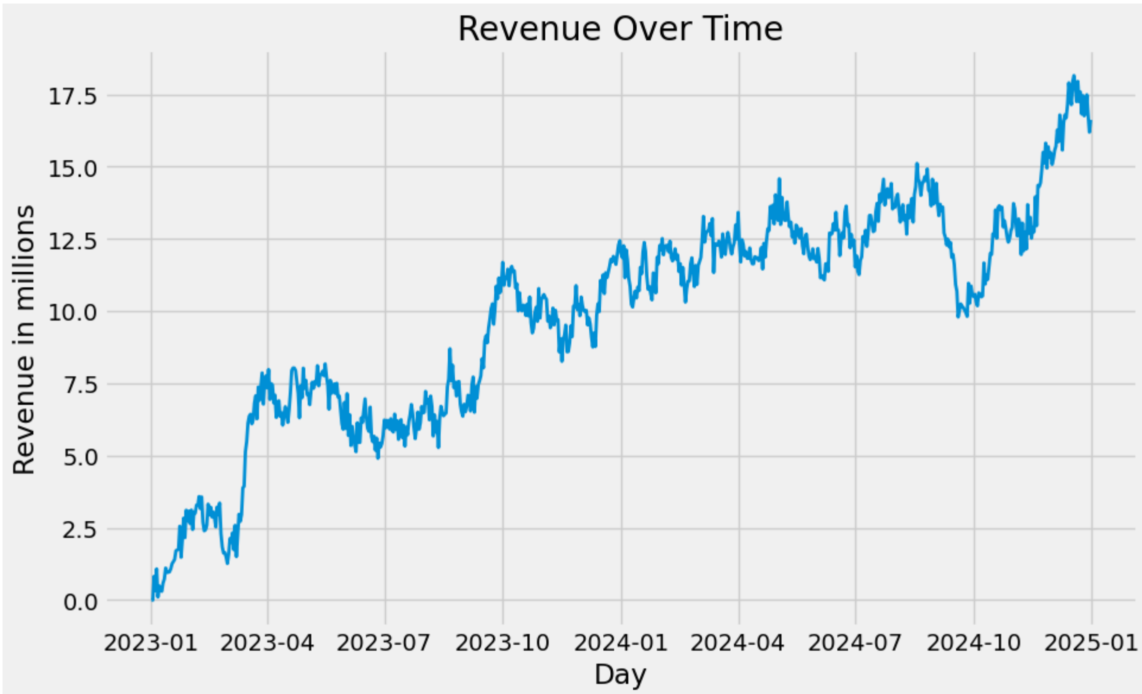
Arima Requirements:

ARIMA time series models operate under several important assumptions.

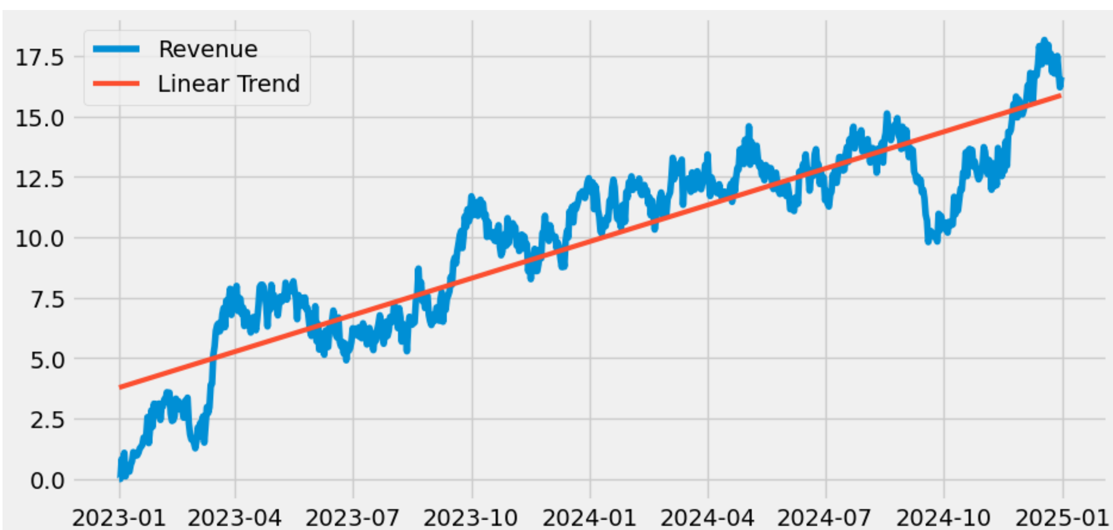
1. The data should be **stationary**, meaning the **mean** and **variance** remain relatively stable over time without strong trends or seasonality. If the data is not stationary, techniques such as differencing may be used to stabilize the series
2. Data should be univariate, meaning the model forecast a single variable using its past values. In this analysis, the daily revenue is the variable being modeled.
3. The model assumes that past observations help predict future observations, which is known as Autocorrelation. This relationship can be evaluated using **ACF** and **PACF** plots.
4. The model assumes that external conditions remain relatively consistent during the observed period, significant events or operational changes may reduce forecasting accuracy because of outside influence.

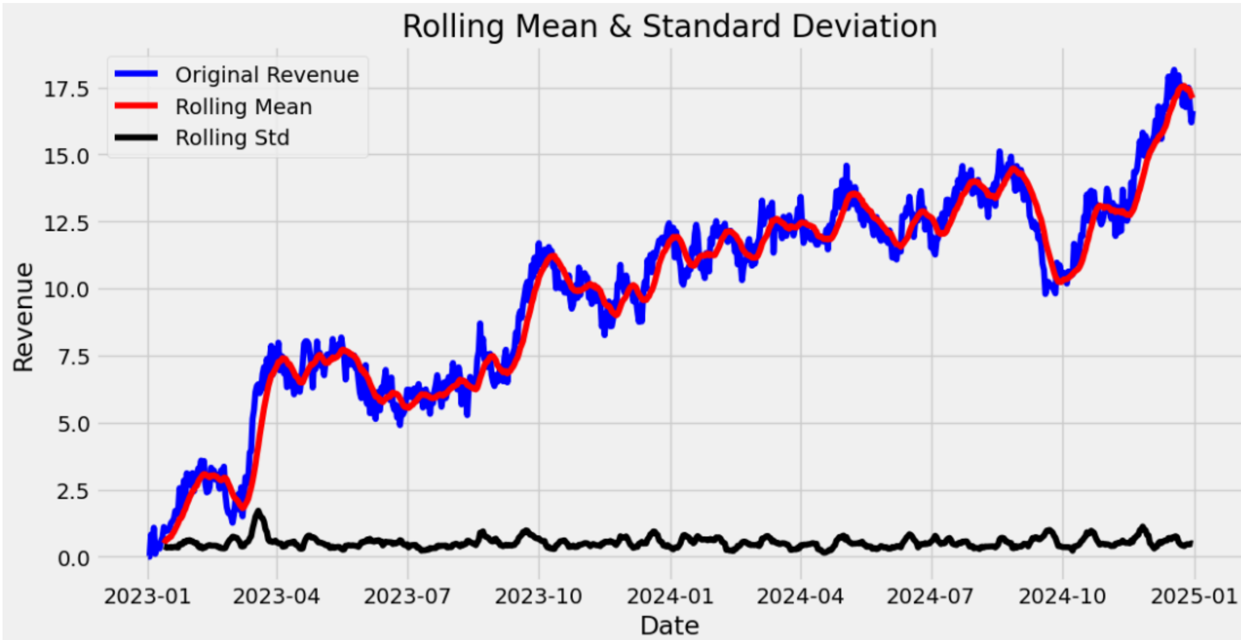
Data Process:

Time series index is initialized on 2023, January 1st as there was no referential date listed.



With trendline:





The original dataset contains **731** observations, with each row representing a single day alongside the corresponding daily revenue in **millions** of U.S. dollars. In the raw dataset, time was represented by a “Day” column containing integers ranging from 1 to 731, indicating the number of days since the beginning of data collection.

To improve interpretability, the “Day” column was converted into a datetime format and replaced with a “Date” column. Because the data dictionary did not specify an actual start date for the data collection period, January 1, 2023 was selected as the starting date to represent approximately two years of daily observations.

The time series progresses in one day increments beginning on January 1, 2023 and ending on December 31, 2024. No gaps or missing time steps were identified in the sequence, as all 731 observations are accounted for in the dataset.

The time series initially appears to be non-**stationary** based on visual inspection of both the revenue plot and the rolling statistics graph. The revenue data shows a clear upward trend over time, indicating the mean of the series increases throughout the observation period rather than remaining constant. In the rolling mean and standard deviation graph, the rolling mean steadily rises instead of remaining flat, further suggesting that the mean changes over time. Although the rolling standard deviation remains relatively stable, the change in rolling mean indicates that the series does not fully satisfy the assumptions of stationarity required for ARIMA modeling.

For quantifiable metrics, the **Augmented Dickey-Fuller (ADF)** test was also performed to statistically evaluate stationarity (Sewell, n.d.).

Null hypothesis = The time series is non-stationary

Alternative hypothesis = The time series is stationary

Using a significance level of **0.05**, the time series failed to reject the null hypothesis at a P-value of **0.32**. This further indicates that the time series is not stationary and requires preprocessing techniques such as differencing before **ARIMA** modeling can be performed.

```
#Do DF test to check if the data is stationary
#Test statistic needs to be more negative than critical value
#p value must be below 0.05 to reject null hypothesis aka state of non stationary


adf_test = adfuller(df['Revenue'])
print("P-Value: ", adf_test[1])

P-Value:  0.32057281507939783
```

Data Preparation Steps:

The data was prepared and analyzed using the following steps:

1. Required libraries and the dataset were imported into Python for analysis.

2. Initial exploratory data analysis was performed to better understand the structure and distribution of the revenue data.

3. The dataset was checked for missing values, outliers, and overall data quality issues before modeling.

4. The dataset was converted into a time series by assigning a daily datetime index to organize the observations chronologically (**2023-01-01 to 2024-12-31**).

5. Visual analysis was performed using revenue plots, rolling mean plots, rolling standard deviation plots, and a trend line to identify patterns such as trends, seasonality, and changing variance (Sewell, n.d.).

6. The stationarity of the series was evaluated through visual inspection to determine whether the mean and variance remained relatively constant over time.

7. The **Augmented Dickey-Fuller (ADF)** test was then performed using a significance level of **0.05** to statistically evaluate stationarity through the p-value for a quantifiable metric.

8. Because the original series was determined to be non-stationary, the data was differenced to stabilize the mean and prepare the series for **ARIMA** modeling while preserving the original dataset for comparison and forecasting purposes (Aly, n.d.).

9. The original dataset containing 731 daily observations was split into a training set of 584 observations (80%) and a testing set of 147 observations (20%). The training set was used to fit the ARIMA model, while the testing set was reserved for evaluating model performance on unseen data. Because time series data are chronological, the **shuffle argument is set to False** in order to retain the chronological order, so the earlier 80% data is used for fitting the model and the later last 20% of the data is the holdout testing data. Test_size = 0.2 = 20%. The time series is initialized to start at 2023-01-01 as seen in the screenshot, and the last 20% of the data starts at 2024-08-07 where testing begins.

   An 80/20 train-test split was selected because it provided sufficient historical observations to train the ARIMA model while retaining a substantial holdout set for model evaluation. The training dataset contains 584 daily observations spanning from 2023-01-01 through 2024-08-06, allowing the model to learn trends, recurring patterns, and revenue behavior across multiple business quarters. The testing dataset contains 147 observations, providing a meaningful out-of-sample period for evaluating forecasting performance on unseen data while preserving the chronological structure required for time series analysis.

```python
#Split the data, Original data

train, test = train_test_split(df, test_size = 0.2, shuffle = False, random_state = 1)
print(train)
print(test)
```

```
              Revenue
2023-01-01   0.000000
2023-01-02   0.000793
2023-01-03   0.825542
2023-01-04   0.320332
2023-01-05   1.082554
...               ...
2024-08-02  13.938920
2024-08-03  14.052184
2024-08-04  13.520478
2024-08-05  13.082643
2024-08-06  13.504886

[584 rows x 1 columns]
              Revenue
2024-08-07  13.684826
2024-08-08  13.152903
2024-08-09  13.310290
2024-08-10  12.665601
2024-08-11  13.660658
...               ...
2024-12-27  16.931559
2024-12-28  17.490666
2024-12-29  16.803638
2024-12-30  16.194813
2024-12-31  16.620798

[147 rows x 1 columns]
```

```
[849]:
#Train Date start and end

print(train)
```

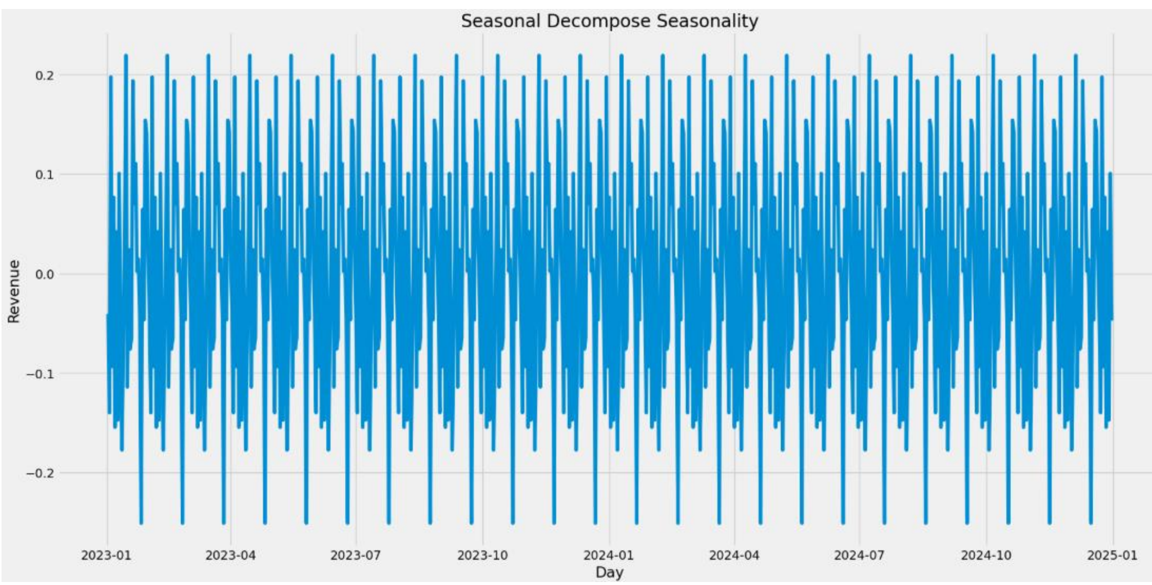
```
              Revenue
2023-01-01   0.000000
2023-01-02   0.000793
2023-01-03   0.825542
2023-01-04   0.320332
2023-01-05   1.082554
...               ...
2024-08-02  13.938920
2024-08-03  14.052184
2024-08-04  13.520478
2024-08-05  13.082643
2024-08-06  13.504886

[584 rows x 1 columns]
```

```
[851]:
#Test Date start and end
print(test)
```

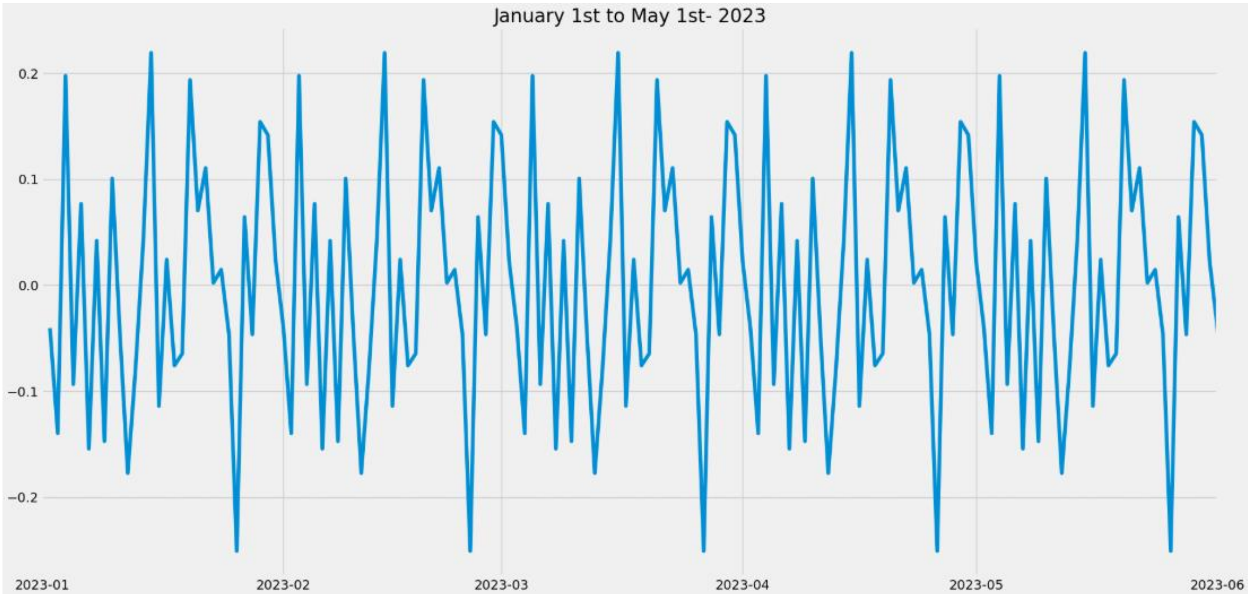
```
              Revenue
2024-08-07  13.684826
2024-08-08  13.152903
2024-08-09  13.310290
2024-08-10  12.665601
2024-08-11  13.660658
...               ...
2024-12-27  16.931559
2024-12-28  17.490666
2024-12-29  16.803638
2024-12-30  16.194813
2024-12-31  16.620798
```

10. After model evaluation, the selected ARIMA(1,1,0) **Final** model was retrained using the entire original, non-differenced revenue dataset. This was done because the train/test split was used for evaluation, while the final forecast should use all available historical observations. The final model was then used to forecast the next 147 days beyond the end of the dataset, matching the same number of days as the testing set for the initial Arima. The forecast was also graphed with confidence intervals to show the expected range of uncertainty over the future forecast period.

The **seasonal decomposition plot** indicates the presence of a seasonal component within the differenced revenue series. Using a **30-day seasonal period**, the component displays repeating cyclical patterns throughout the timeframe, indicating the existence of recurring monthly behavior within the revenue data. The consistent oscillating line over time suggests that **seasonality remains present** even after differencing was applied to remove stationarity **ARIMA modeling**.

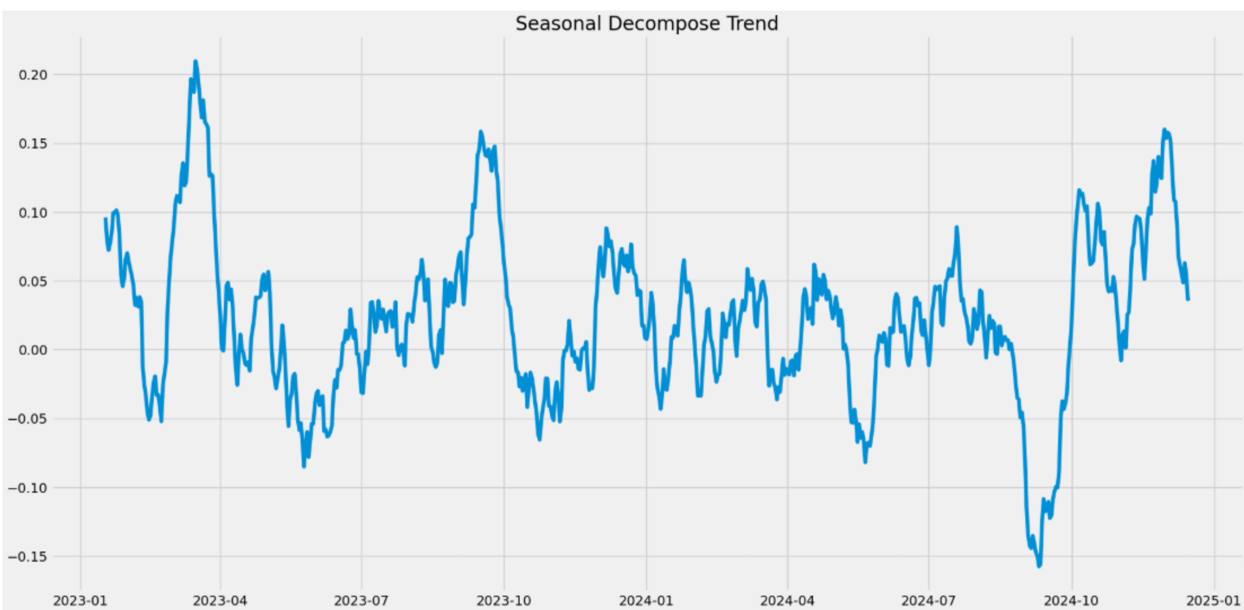


With magnification and subsetting the time series for the first 6 months of the data (2023, Jan 1st to May 1st), clearly shows seasonality.



Trends:

The **decomposed trend component** of the **differenced revenue** series **fluctuates** around **zero** without displaying a sustained upward or downward directional movement. Unlike the original revenue series, which showed a clear upward trend, the differenced series appears more stable over time. This suggests that differencing successfully reduced the long-term trend and improved stationarity for ARIMA modeling.

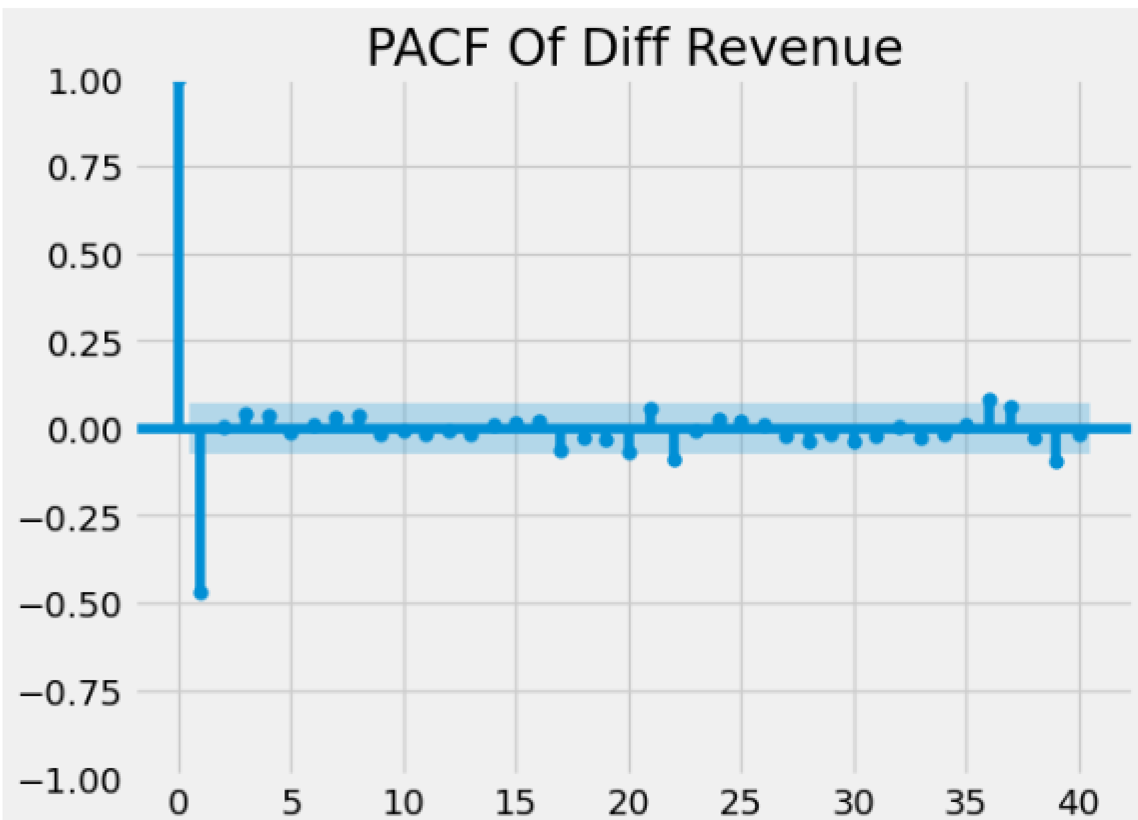


Autocorrelation:

Both **PACF** and **ACF** are used to determine **p** and **q**, **PACF** to determine **p** and **ACF** to determine **q** of the **p,d,f** inputs for **ARIMA**.

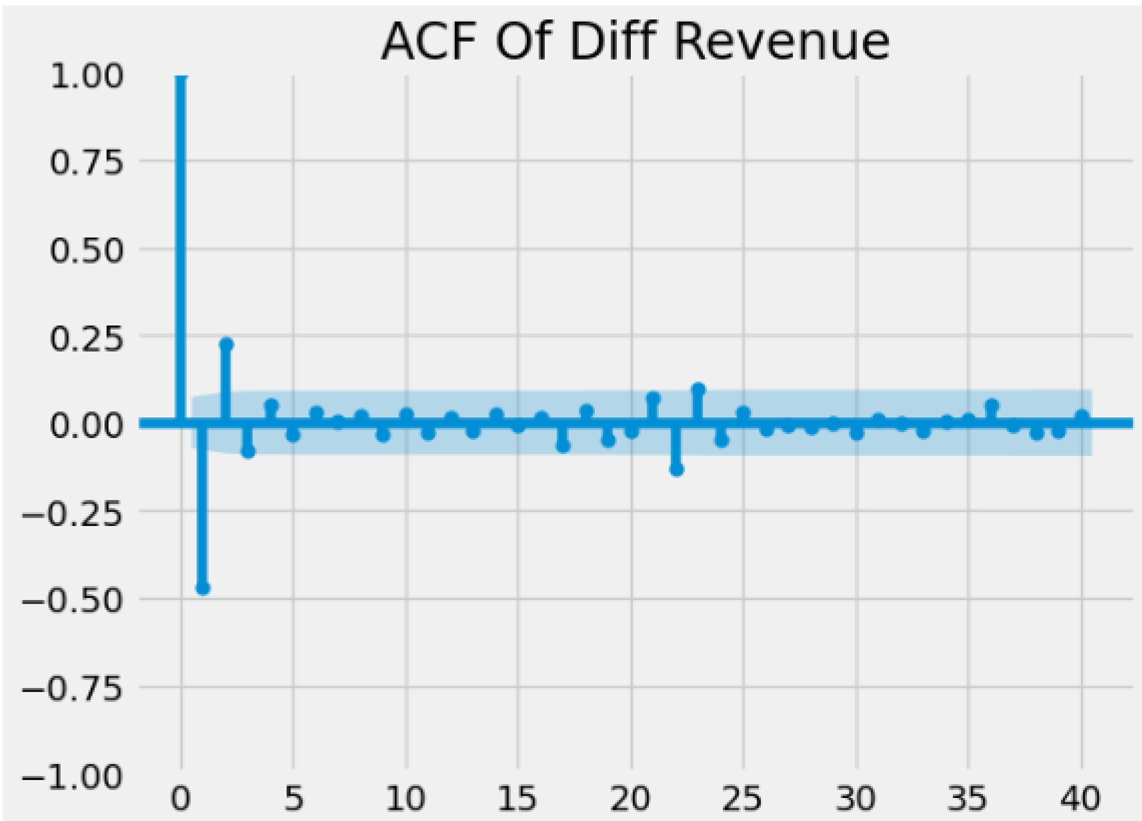
PACF:

The **PACF** plot of the differenced revenue series shows a strong negative spike at **lag 1** followed by values that mostly remain within the confidence interval bounds. This suggests that the autoregressive effect is concentrated in the first lag, supporting a potential **AR term of p = 1**. The lack of significant spikes at later lags indicates that the differencing helped reduce autocorrelation within the series.



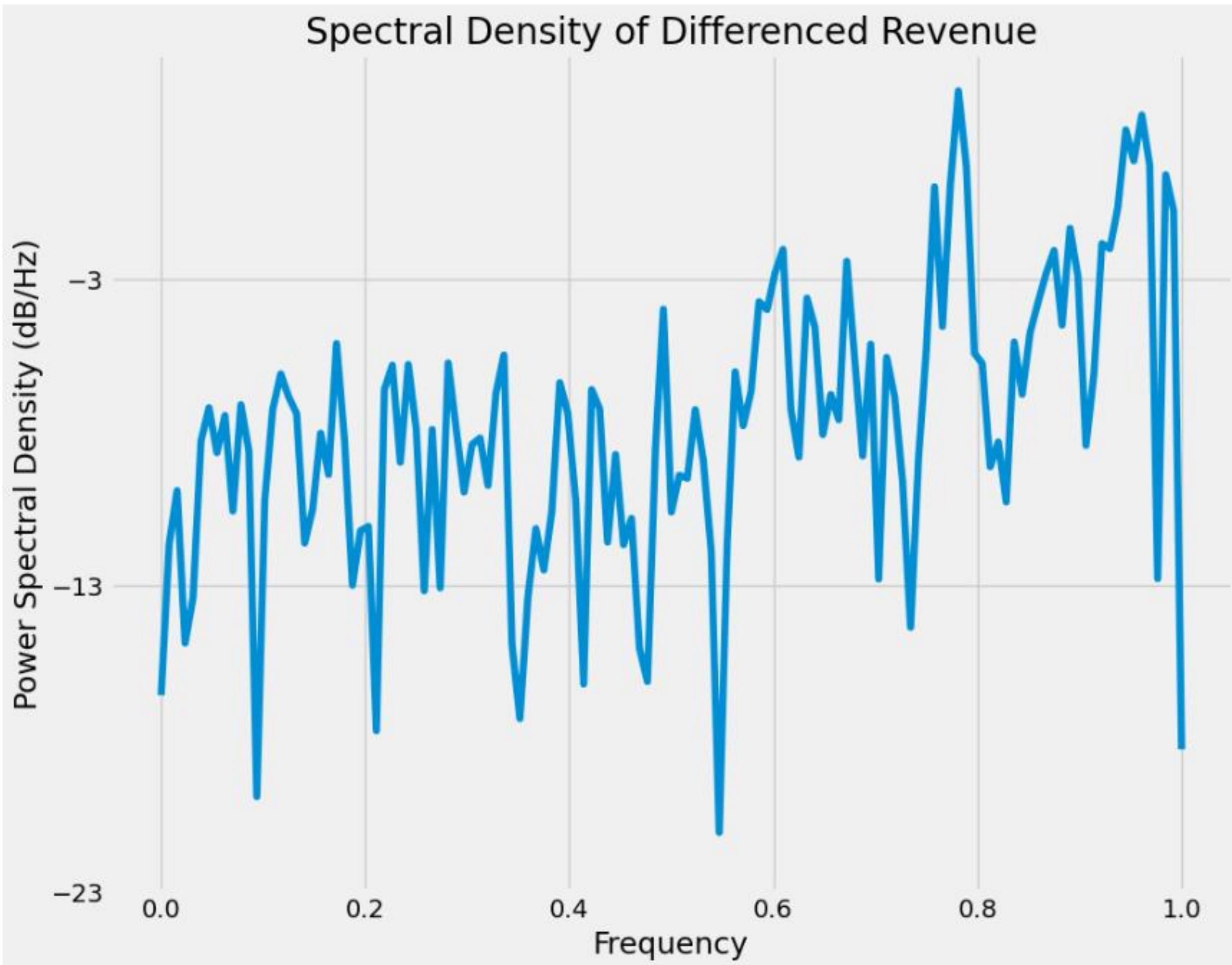
ACF:

The **ACF** plot of the differenced revenue series shows a strong negative spike at **lag 1** and a smaller positive spike at **lag 2**, while most remaining lags stay within the confidence interval bounds. This suggests that the moving average effect is mainly concentrated within the first few lags, supporting a low-order **MA** component for **ARIMA modeling**. The rapid decline in autocorrelation after the first few lags also suggests that differencing helped make the series more stationary by reducing the relationship between past observations over time.



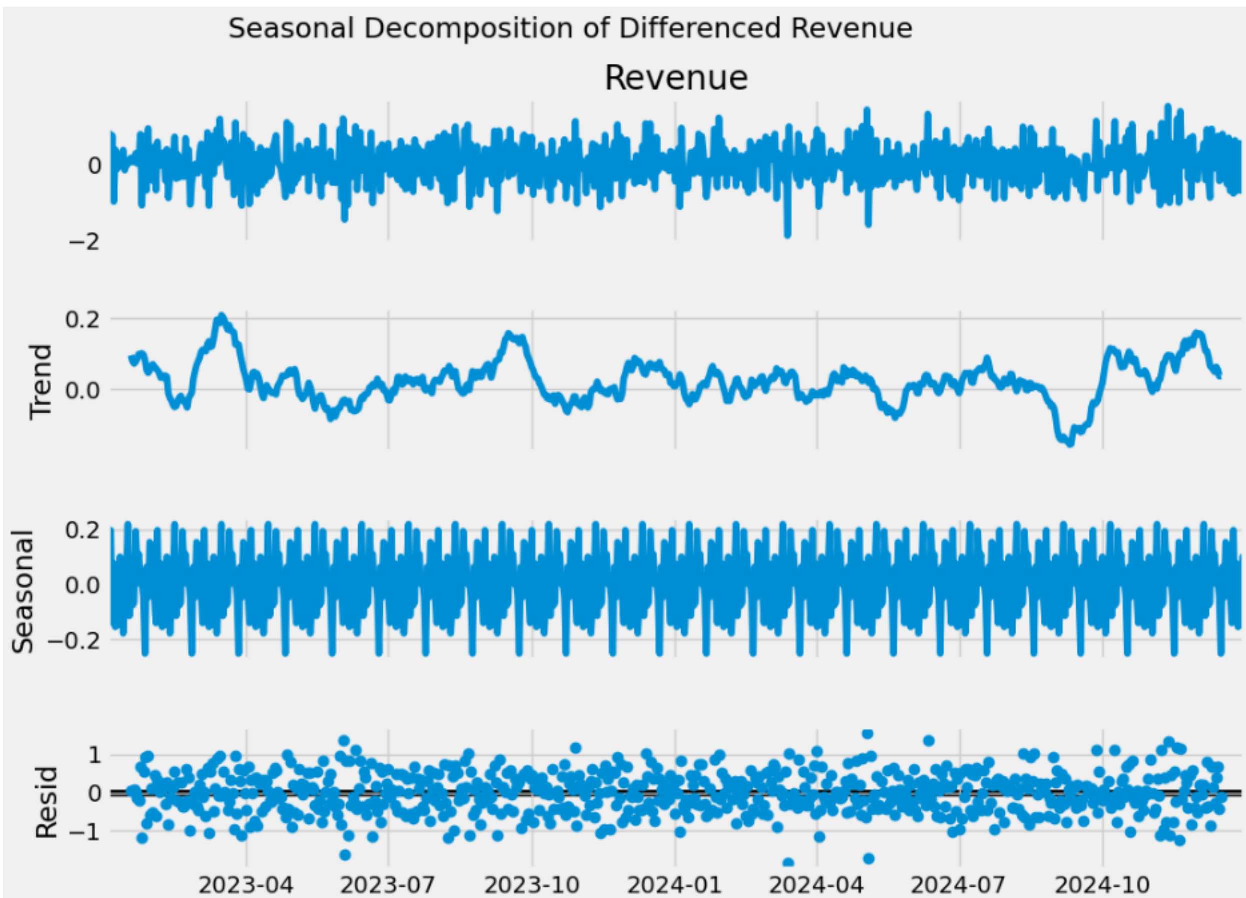
Spectral Density:

The **spectral density plot** of the differenced revenue series shows fluctuations across multiple frequencies, suggesting a weak presence of some cyclical or seasonal behavior within the data. However, the absence of a single dominant peak indicates that the seasonal patterns are relatively **weak** and **mixed** with random variation.



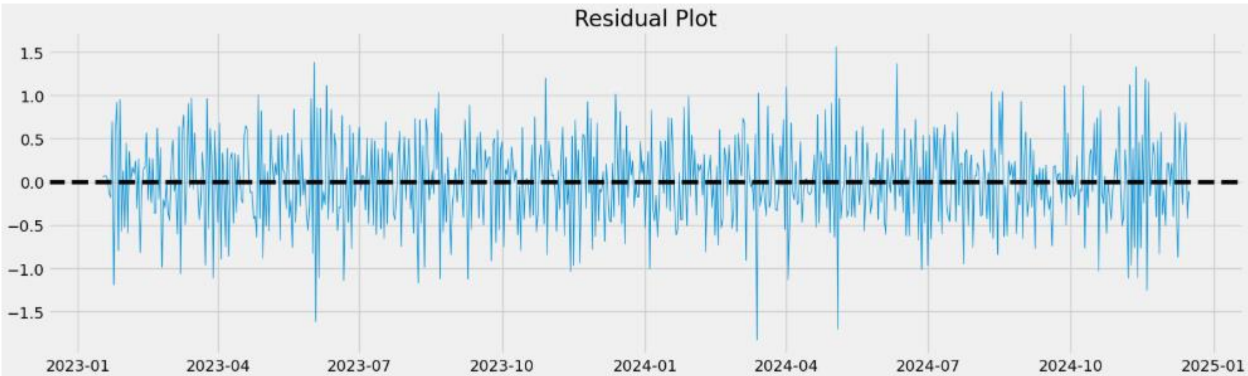
Decomposed Time Series:

The **decomposed time series** separates the **differenced revenue data** into **trend**, **seasonal**, and **residual components**. The revenue graph shows the data after being differenced. The trend component appears relatively stable with minor fluctuations over time, with no consistent upward or downward trend. While the seasonal component shows repeating cyclical patterns that indicate the presence of seasonality within the series. The residual component appears randomly distributed around zero without a clear trend, suggesting that much of the systematic structure was successfully captured through decomposition (Sewell, n.d.).



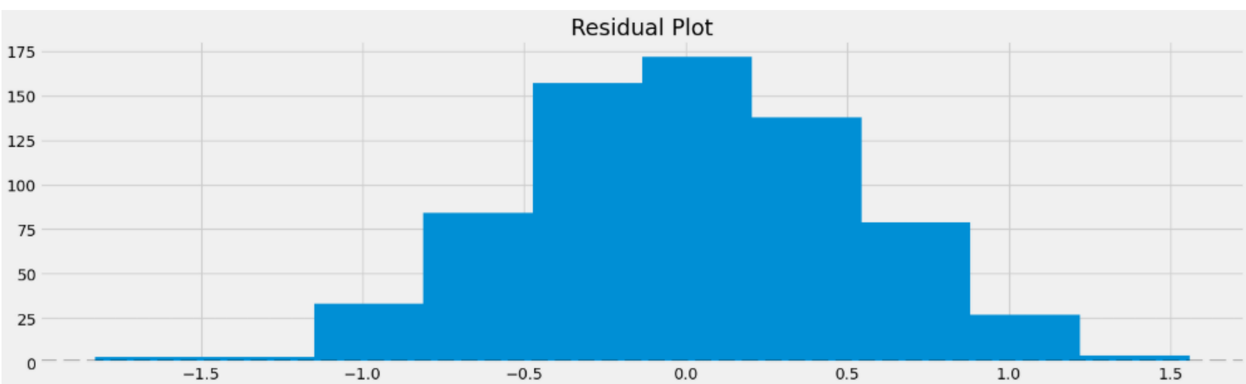
Confirmation of the lack of trends in the residuals of the decomposed series:

The **residual plot** shows that the residual values fluctuate around zero without a clear upward or downward trend over time. This confirms that the differencing successfully removed most of the trend.



Histogram version:

Residuals hover around zero as a normal distribution with no randomness or spikes at the beginning or end.



Based on the **differenced revenue time series**, the initial **ARIMA** model was manually identified as **ARIMA (1,1,2)** through the analysis of **PACF, ACF plots** and **differencing** once to make the time series stationary. The **PACF plot** showed a strong negative spike at lag 1 followed by values that mostly remained within the confidence interval bounds, suggesting a p = 1. The ACF plot displayed significant autocorrelation within the first few lags before crossing zero, suggesting a possible moving average component of q = 2. The differencing parameter of d = 1 was selected because differencing once helped achieve a P-value of less than 0.05, making it significant and stationary.

After the initial manual model selection, the **AutoARIMA** function was established to compare multiple **ARIMA** configurations using **AIC** (Akaike Information Criterion) (Aly, n.d.). Although the manually selected **ARIMA(1,1,2)** model was reasonable based on the visual interpretation of the **ACF** and **PACF** plots, **AutoARIMA** identified ARIMA**(1,1,0)** as the optimal model due to its lower **AIC** value and **reduced complexity**. Therefore **ARIMA(1,1,0)** was selected as the final forecasting model parameters for this analysis. ARIMA(1,1,1) was also explored as to bridge the gap between the first two models.

AutoArima:

```
#Do autoArima
stepwise_fit = auto_arima(df['Revenue'], trace =True)
stepwise_fit.summary()
```

```
Performing stepwise search to minimize aic
 ARIMA(2,1,2)(0,0,0)[0] intercept   : AIC=987.305, Time=0.36 sec
 ARIMA(0,1,0)(0,0,0)[0] intercept   : AIC=1162.819, Time=0.03 sec
 ARIMA(1,1,0)(0,0,0)[0] intercept   : AIC=983.122, Time=0.04 sec
 ARIMA(0,1,1)(0,0,0)[0] intercept   : AIC=1019.369, Time=0.05 sec
 ARIMA(0,1,0)(0,0,0)[0]             : AIC=1162.139, Time=0.03 sec
 ARIMA(2,1,0)(0,0,0)[0] intercept   : AIC=985.104, Time=0.07 sec
 ARIMA(1,1,1)(0,0,0)[0] intercept   : AIC=985.106, Time=0.04 sec
 ARIMA(2,1,1)(0,0,0)[0] intercept   : AIC=986.045, Time=0.24 sec
 ARIMA(1,1,0)(0,0,0)[0]             : AIC=984.710, Time=0.02 sec

Best model:  ARIMA(1,1,0)(0,0,0)[0] intercept
Total fit time: 0.888 seconds
```

ARIMA(1,1,2):

AIC	986.818
BIC	1005.190

ARIMA(1,1,0)

AIC	984.710
BIC	993.896

ARIMA(1,1,1)

AIC	986.652
BIC	1000.431

Final model:

```
#Make model

model = ARIMA(train['Revenue'], order= (1,1,0))
model = model.fit()
model.summary()
```

SARIMAX Results

Dep. Variable:	Revenue	No. Observations:	584
Model:	ARIMA(1, 1, 0)	Log Likelihood	-385.018
Date:	Sun, 24 May 2026	AIC	774.035
Time:	14:58:59	BIC	782.772
Sample:	01-01-2023	HQIC	777.441
	- 08-06-2024		
Covariance Type:	opg		

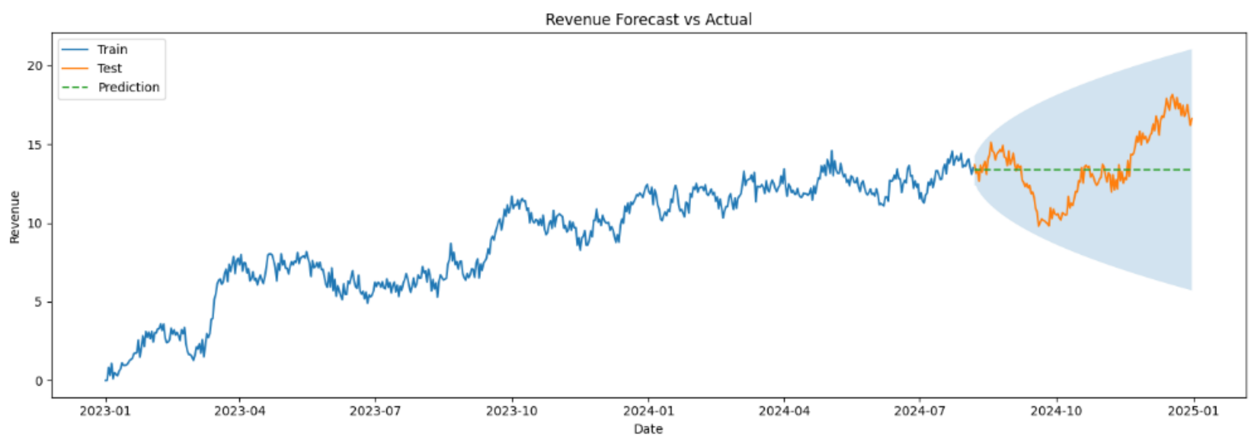
	coef	std err	z	P>|z|	[0.025	0.975]
ar.L1	-0.4578	0.036	-12.618	0.000	-0.529	-0.387
sigma2	0.2193	0.014	15.954	0.000	0.192	0.246

Ljung-Box (L1) (Q):	0.01	Jarque-Bera (JB):	1.81
Prob(Q):	0.91	Prob(JB):	0.40
Heteroskedasticity (H):	0.97	Skew:	-0.07
Prob(H) (two-sided):	0.81	Kurtosis:	2.77

Forecasting:

The forecast visualization compares the training data, testing data and ARIMA model predictions for the revenue time series. The blue line represents the **historical** training data used to fit the **ARIMA(1,1,0)** model, while the orange line represents the **unseen** testing data used for forecast evaluation. The **dashed green line** represents the **forecasted** revenue values generated by the model, and the **blue** shaded region represents the **confidence interval**, showing the increasing uncertainty further as time increases (Aly, n.d.)

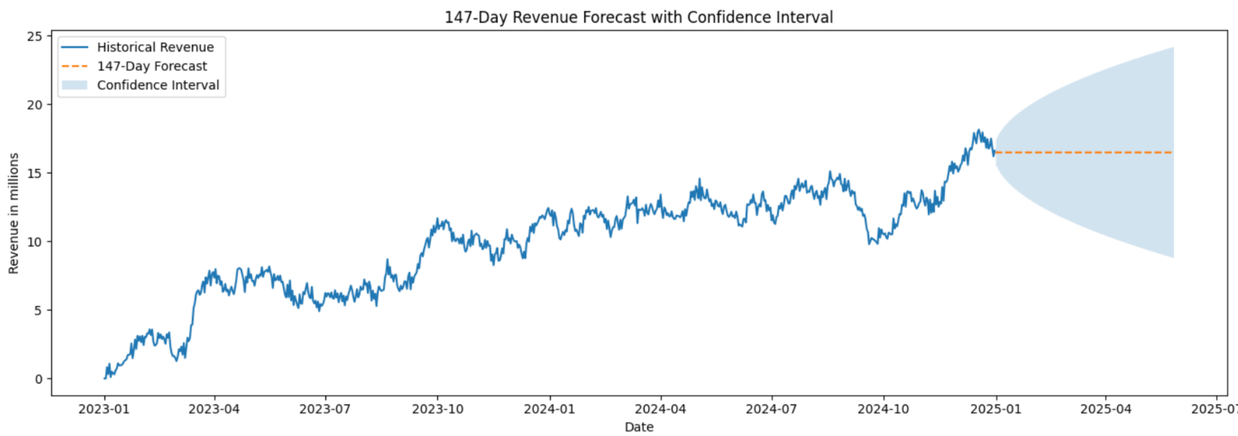
The forecast produced a relatively stable prediction line across the testing period, suggesting that the **ARIMA** model estimated the **future revenue** to remain near the **average level** (around 13.5/14 million in revenue) observed at the end of the training data. The data was split into an 80/20 train/test split. Although the forecast did not capture every fluctuation within the testing data, much of the observed revenue remained within the confidence interval bounds.



After evaluating model performance using the 80/20 train-test split, the selected ARIMA(1,1,0) model was retrained using the entire original revenue dataset containing 731 observations. Retraining on the full dataset allowed the model to utilize all available historical information before generating the final forecast.

The retrained model was then used to forecast the next 147 days beyond the end of the observed dataset (2025-01-01 through 2025-05-27). The forecast produced a relatively stable revenue projection near 16.5 million, suggesting that future revenue is expected to remain near the average level observed at the end of the historical period. The confidence interval widens as the forecast horizon extends further into the future, indicating increasing uncertainty in longer-term predictions.

A 147-day forecast horizon was selected to match the size of the testing dataset used during model evaluation, providing consistency between the evaluation period and the final future forecast. This horizon also provides approximately five months of projected revenue while avoiding excessive uncertainty associated with very long-range forecasts.



ARIMA(1,1,0) model is initialized

```
#Make model

model = ARIMA(train['Revenue'], order= (1,1,0))
model = model.fit()
model.summary()
```

SARIMAX Results

Dep. Variable:	Revenue	**No. Observations:**	584
Model:	ARIMA(1, 1, 0)	**Log Likelihood**	-385.018
Date:	Sun, 24 May 2026	**AIC**	774.035
Time:	14:58:59	**BIC**	782.772
Sample:	01-01-2023	**HQIC**	777.441
	- 08-06-2024		
Covariance Type:	opg		

	coef	std err	z	P>|z|	[0.025	0.975]
ar.L1	-0.4578	0.036	-12.618	0.000	-0.529	-0.387
sigma2	0.2193	0.014	15.954	0.000	0.192	0.246

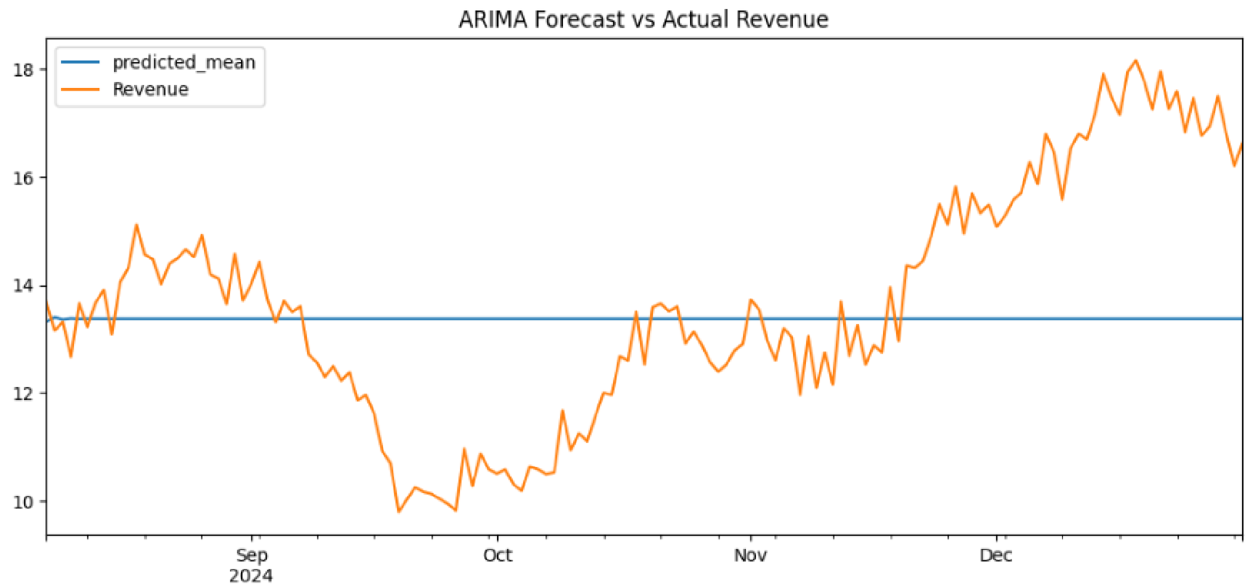
Ljung-Box (L1) (Q):	0.01	**Jarque-Bera (JB):**	1.81
Prob(Q):	0.91	**Prob(JB):**	0.40
Heteroskedasticity (H):	0.97	**Skew:**	-0.07
Prob(H) (two-sided):	0.81	**Kurtosis:**	2.77

•[1138…

```python
# Make predictions on test set

start = len(train)
end = len(train) + len(test) - 1

pred = model.predict(start=start, end=end, typ='levels')

# Match prediction index to test dates
pred.index = test.index

print(pred)
```

```
2024-08-07    13.311586
2024-08-08    13.400077
2024-08-09    13.359566
2024-08-10    13.378112
2024-08-11    13.369622
                ...
2024-12-27    13.372288
2024-12-28    13.372288
2024-12-29    13.372288
2024-12-30    13.372288
2024-12-31    13.372288
Freq: D, Name: predicted_mean, Length: 147, dtype: float64
```

```
#Get Mean Line (Sewell)

pred.plot(legend=True, figsize=(12,5))
test['Revenue'].plot(legend=True)

plt.title("ARIMA Forecast vs Actual Revenue")
plt.show()
```



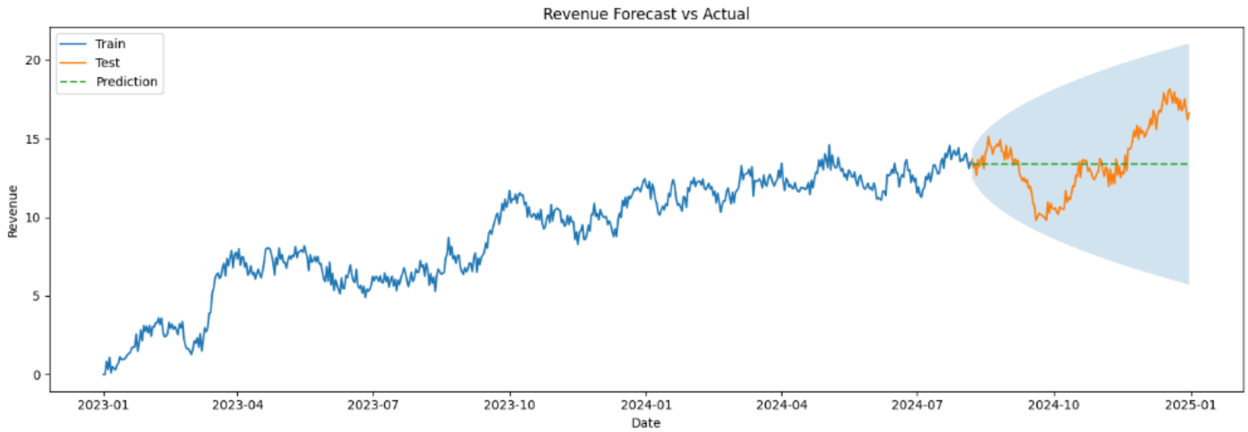
This code generates forecast on the test dataset, extracts predictions, aligns the forecast dates with the testing data and calculate forecasting error metrics including MAE, MSE< and RMSE to evaluate model performance

```
# Forecast on test set
fc_test = model.get_forecast(steps=len(test))
fc_mean = fc_test.predicted_mean
fc_ci = fc_test.conf_int()

# Match forecast index with test index
fc_mean.index = test.index
fc_ci.index = test.index

mae = mean_absolute_error(test['Revenue'], fc_mean)
mse = mean_squared_error(test['Revenue'], fc_mean)
rmse = np.sqrt(mse)

print("MAE:", mae)
print("MSE:", mse)
print("RMSE:", rmse)
```

```
MAE: 1.738607878329873
MSE: 4.737181349875626
RMSE: 2.1765066850059585
```

Code to visualize predictions (Aly, n.d.)

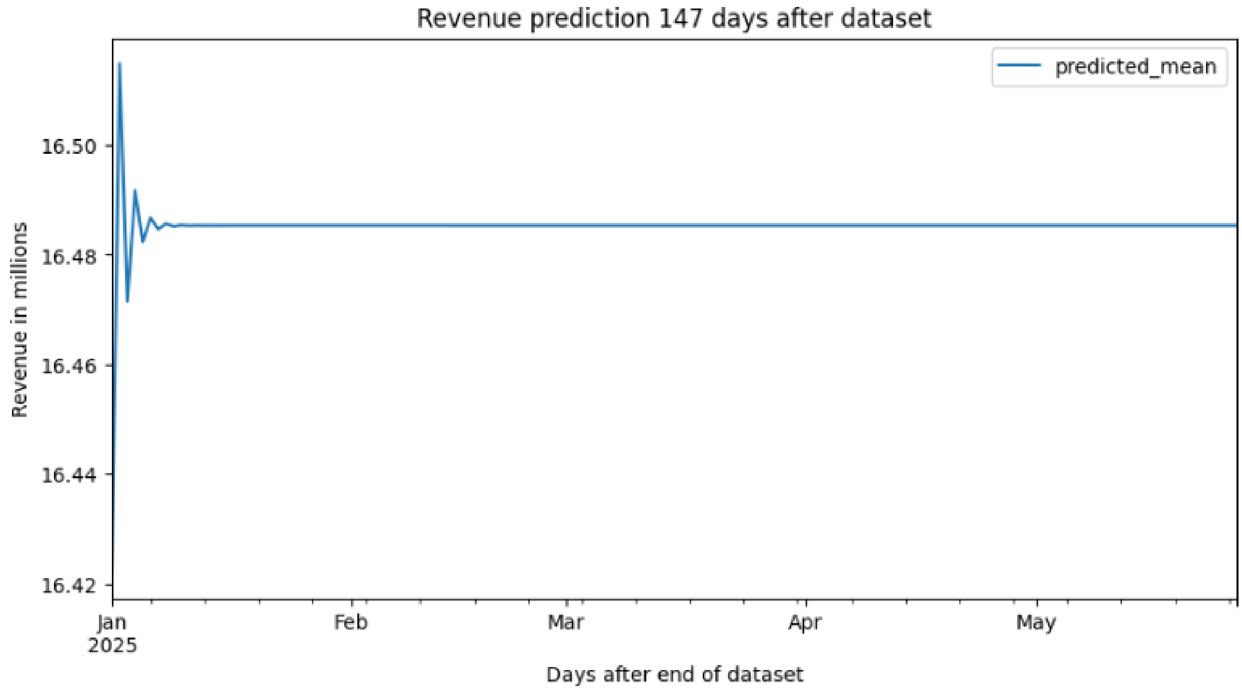
```
plt.figure(figsize=(14,5))

plt.plot(train['Revenue'], label='Train')
plt.plot(test['Revenue'], label='Test')
plt.plot(fc_mean, label='Prediction', linestyle='--')

plt.fill_between(test.index, fc_ci.iloc[:,0], fc_ci.iloc[:,1], alpha=0.2)

plt.title("Revenue Forecast vs Actual")
plt.xlabel("Date")
plt.ylabel("Revenue")
plt.legend()

plt.tight_layout()
plt.show()
```

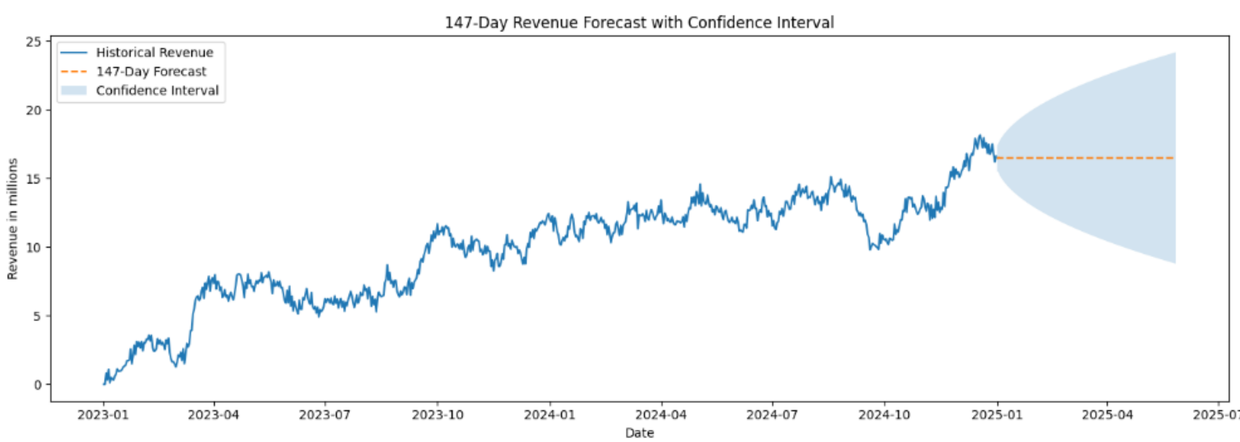


Final Model: 147 days of forecast after the dataset ends

```
# Forecast the next 147 days after the dataset ends
#Revise

# Train ARIMA model on the full dataset
final_model = ARIMA(df['Revenue'], order=(1,1,0))
final_model = final_model.fit()

future_dates = pd.date_range(start='2025-01-01', periods=147, freq='D')

future_forecast = final_model.get_forecast(steps=147)


future_values = future_forecast.predicted_mean

future_values.index = future_dates

print(future_values)
```

```
2025-01-01    16.422010
2025-01-02    16.514776
2025-01-03    16.471486
2025-01-04    16.491688
2025-01-05    16.482261
                ...
2025-05-23    16.485260
2025-05-24    16.485260
2025-05-25    16.485260
2025-05-26    16.485260
2025-05-27    16.485260
Freq: D, Name: predicted_mean, Length: 147, dtype: float64
```


Revenue prediction 147 days after dataset
predicted_mean
Revenue in millions
16.50
16.48
16.46
16.44
16.42
Jan
2025
Feb
Mar
Apr
May
Days after end of dataset


147-Day Revenue Forecast with Confidence Interval
Historical Revenue
147-Day Forecast
Confidence Interval
Revenue in millions
25
20
15
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Date

Result Summary:

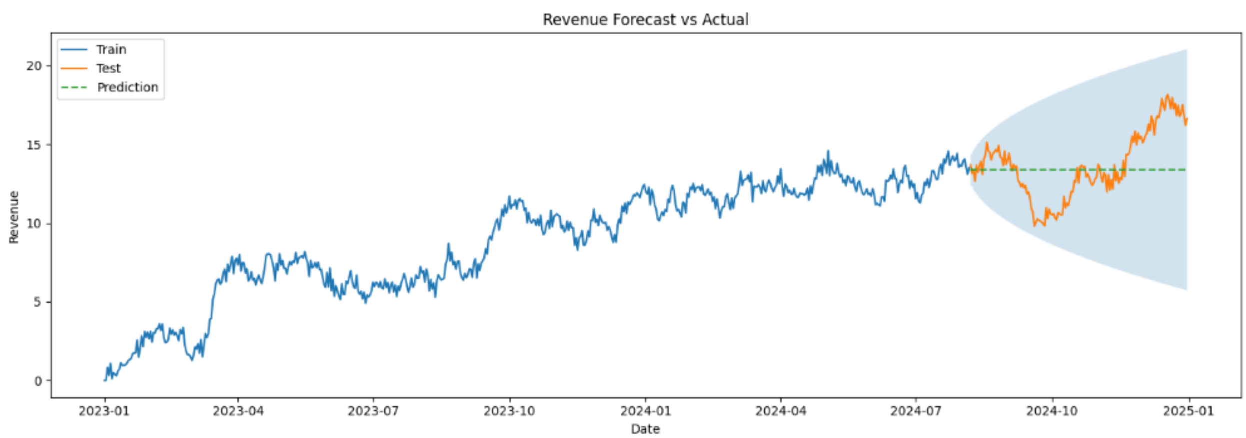
The **ARIMA** model selection process involved **evaluating stationarity**, **autocorrelation**, and model performance metrics (**AIC**). After **differencing** was applied to stabilize the mean and reduce trend within the revenue series (**ADfuller**), the **PACF** and **ACF** plots were evaluated to estimate the **p, d, q** parameters. Initial manual analysis suggested an **ARIMA (1,1,2).** However, **AutoARIMA** was later used to compare multiple **ARIMA** configurations using **AIC**(Akaike Information Criterion) values, which identified ARIMA**(1,1,0)** as the optimal model due to its lower complexity and improved model performance.

The prediction interval spans **147** days following the **584** days of training data, based on an **80/20** train-test split of the two-year revenue dataset. This provided adequate historical observations to capture long term trends and recurring behavior across more than four quarters of revenue activity, while reserving the final 20% observations for forecast evaluation on unseen data. As the forecast extenders further into the future, the confidence interval widens over time, indicating increasing uncertainty (Sewell, n.d.).

This forecasting length provided a sufficient number of unseen future observations for evaluating the model performance while avoiding excessively long-term forecast that may reduce reliability due to increasing confidence interval that widens over time, as the model becomes less certain of its prediction capability (Aly, n.d.). Additionally, the dataset initially did not have an established date, being generous with the train split would allow the testing data to cover at least 4 quarters of a financial fiscal year.

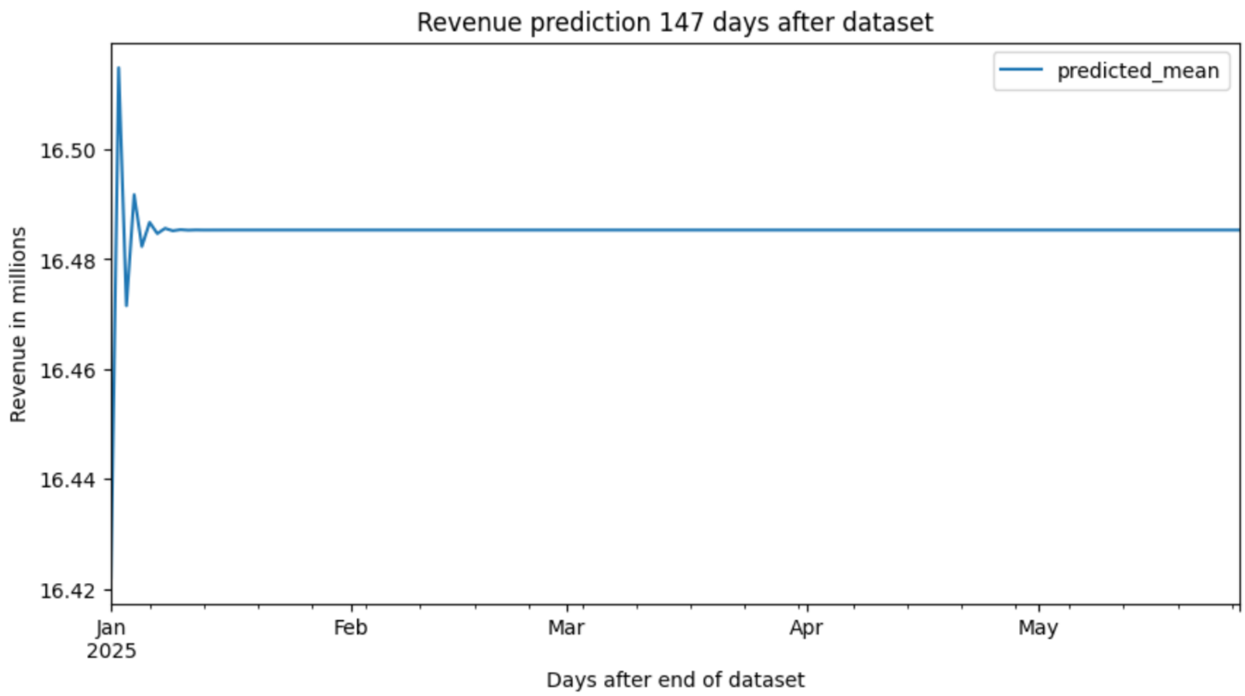
The ARIMA model was evaluated using **RMSE**(Root Mean Squared Error) specifically as it is highly interpretable and remains in the same unit as the data (Revenue in millions). The **RMSE** of the model is 2.17, meaning that the forecasted revenue values were typically within plus or minus **2.17 million dollars** of the actual observed revenue values during the testing period. Compared to the average testing revenue of **13.64 million dollars**, this indicates that the **ARIMA** model produced reasonably accurate forecast while still showing some forecasting error during periods of higher volatility and upward movement within the testing dataset.

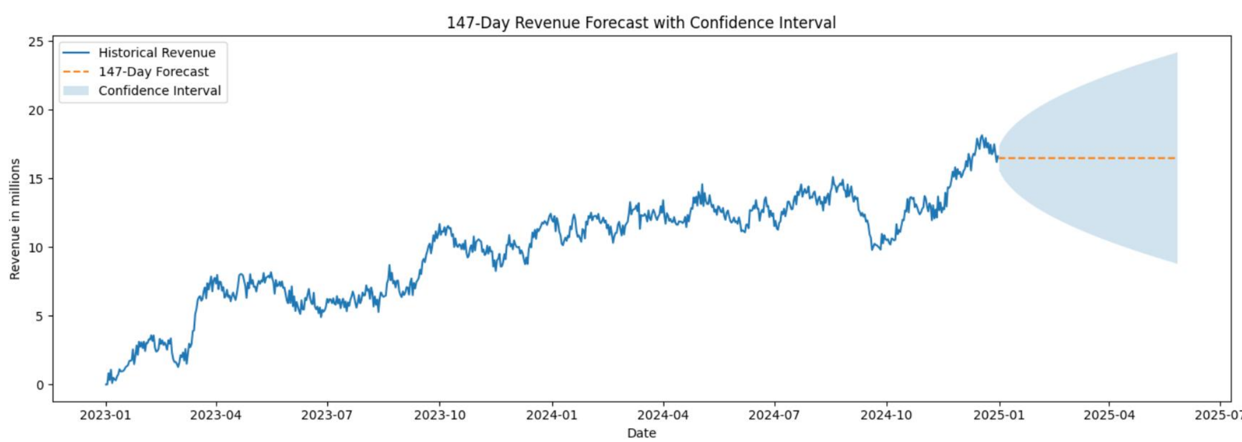
The visualization compares the **ARIMA**(1,1,0) forecast against the testing dataset. The blue line represents the historical training data used to fit the model, while the orange represents the unseen testing data used to evaluate the forecasting performance. The dashed orange line represents the forecast generated by the **ARIMA** model, and the shaded region represents the confidence interval, which widens over time as uncertainty increases further into the forecast horizon. The forecast produced a relatively stable prediction near the average revenue level observed at the end of the training dataset. Although the model did not capture all fluctuations, peaks, and dips present within the testing data, much of the observed testing data remained within the confidence interval bounds. This suggests that the **ARIMA** model was reasonably effective at estimating the general behavior and overall level of the revenue series while acknowledging the increasing uncertainty in longer-term predictions.



The final model is a 147-day future forecast visualization after the dataset ends is initialized, to further support this observation. The model quickly switches to a nearly constant prediction value of around 16.48 million dollars. This model was trained on the whole dataset (Aly, n.d.).

The 147-day Revenue Forecast with Confidence Interval shows the forecasted data stays within the confidence interval.





Recommended action based on results:

The **ARIMA**(1,1,0) model was first evaluated using the **80/20** train-test split. The in-sample test prediction showed that the model produced a relatively flat prediction line and did not fully capture the short-term upward and downward fluctuations observed in the testing data. Although the model produced acceptable error metrics such as RMSE, the visualization suggests that the model is more effective at estimating the overall mean revenue level and long-term trend than predicting sudden volatility or daily fluctuations.

After evaluation, the final model was retrained using the entire dataset containing **731** observations to create the final **147-day** forecast beyond the observed data. This approach is appropriate because the test split was used for model evaluation, while the final forecast should utilize all available historical information. A **147**-day forecasting horizon was selected because it matches the size of the testing dataset and provides approximately five months of future revenue projections. This forecast horizon provides a meaningful future outlook while avoiding the excessive uncertainty associated with very long-range forecasts.

The final model forecast indicates that revenue is expected to remain relatively stable near 16.5 million dollars throughout the forecast period. While the forecast mean remains nearly constant, the confidence interval widens as the forecast extends further into the future, indicating increasing uncertainty in the projected revenue values. This behavior is expected in time series forecasting because prediction uncertainty accumulates as the forecast horizon increases.

Based on these results, the recommended course of action is to use the ARIMA forecast as a conservative baseline for short- to medium-term revenue planning rather than as a precise prediction of daily revenue changes. The model provides a reasonable estimate of the overall revenue level and direction of future performance, but additional historical data should be collected to improve future model reliability and support stronger revenue forecasting decisions.